

Ex.No.10 - 5G Network Slicing Environment Using MATLAB

Objective – 1 Analyze the Number of Network Slices

MATLAB Code:

```
clc;

clear;

close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1];

figure

bar(count)

% One instance of each network slice

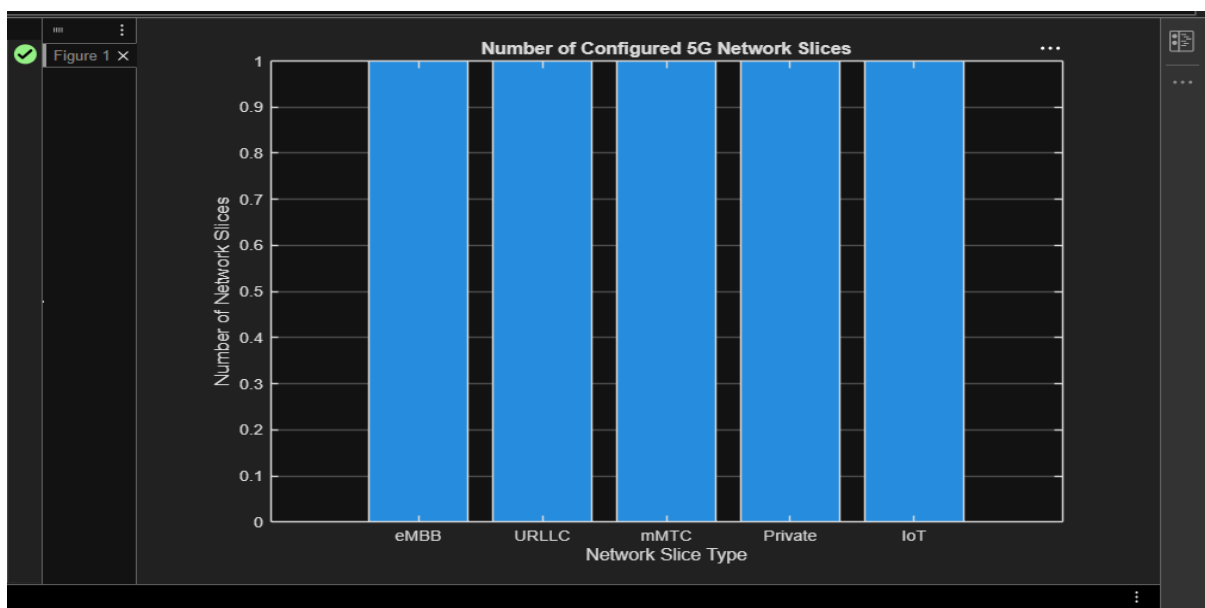
grid on

set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

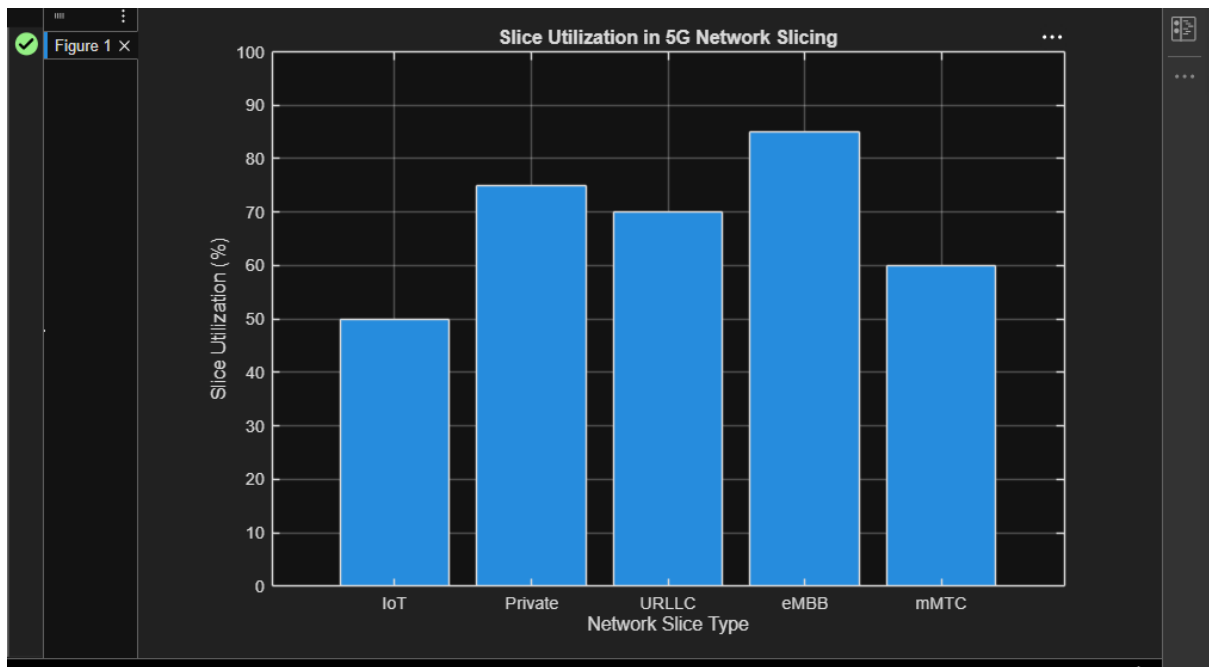
title('Number of Configured 5G Network Slices')
```



Objective – 2 Compare Slice Utilization (%)

MATLAB Code

```
clc;  
clear;  
close all;  
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});  
utilization = [85 70 60 75 50];  
figure;  
bar(slices, utilization);  
grid on;  
xlabel('Network Slice Type');  
ylabel('Slice Utilization (%)');  
title('Slice Utilization in 5G Network Slicing');  
ylim([0 100]);
```



Objective – 3 Analyze End-to-End Latency (ms)

```
clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

figure;

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

grid on;

xticks(slices);

xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});

xlabel('Network Slice Type');

ylabel('End-to-End Latency (ms)');

title('5G Network Slicing: End-to-End Latency');

ylim([0 70]);
```

